

KATUNAYAKE, Sri Lanka

WMO: 434500

Lat: 7.167N

Lon: 79.883E

Elev: 26

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	69.8	71.6	60.7	79.3	85.0	62.8	85.6	81.9	20.7	84.4	18.9	84.0	6.4	40	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	156.7	86.2	79.1	151.3	85.3	78.8	149.8	85.1	45.5	87.9	44.8	87.8	44.2	86.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.8	15.2	66.0	95.0	1.7	1.8	64.8	96.3	63.7	97.4	62.8	98.4	61.5	99.7		
(5)			63.3	82.3	1.7	1.0	62.1	83.0	61.1	83.6	60.1	84.2	58.9	85.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	80.4	81.4	82.6	83.0	83.5	82.8	82.4	82.2	82.0	81.2	80.6	80.3
	DBStd	1.98	1.92	2.06	1.58	1.82	1.88	1.68	1.53	1.57	1.62	1.53	1.46	1.60
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11632	942	880	1009	991	1040	983	1004	999	959	967	919	939
	CDD65	6157	477	460	544	541	575	533	539	534	509	502	469	474
	CDH74	66002	4424	4722	6269	6236	7061	6265	6018	5860	5576	5108	4261	4202
	CDH80	23093	1601	1860	2614	2631	2885	2158	1777	1691	1687	1548	1303	1338
Wind	WSAvg	8.0	9.1	7.6	6.2	6.1	8.6	9.7	9.7	9.7	8.9	6.7	5.8	7.9
Precipitation	PrecAvg	80.0	1.3	2.4	5.2	9.3	12.6	5.5	4.4	2.7	6.2	14.4	10.9	5.0
	PrecMax	137.7	5.7	7.6	10.6	19.6	26.9	12.7	18.0	11.0	13.9	27.6	21.1	20.0
	PrecMin	44.3	0.0	0.0	0.5	2.1	1.0	0.6	0.0	0.0	0.4	2.9	0.5	0.2
	PrecStd	21.5	1.6	2.0	2.9	4.5	7.2	3.4	4.3	2.8	4.1	6.5	5.5	4.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	94.7	93.4	91.8	90.8	88.5	88.1	88.1	88.2	88.2	90.0	91.0
		MCWB	73.2	74.1	75.9	80.5	80.8	80.5	79.7	78.8	78.6	78.2	75.6	74.3
	2%	DB	90.0	91.7	91.2	91.1	89.8	87.9	86.4	87.3	87.6	87.7	88.2	89.3
		MCWB	73.4	73.7	77.8	80.1	80.6	80.3	78.8	78.7	78.6	78.2	76.6	74.7
	5%	DB	89.2	89.8	89.7	89.8	89.2	86.6	86.2	86.2	86.3	86.3	87.5	87.8
		MCWB	73.6	74.8	77.9	79.7	80.5	79.5	78.6	78.4	78.3	78.1	76.9	75.3
	10%	DB	87.6	88.1	88.3	89.2	87.9	86.1	85.7	85.8	85.9	85.9	86.1	86.2
		MCWB	74.0	75.7	77.9	79.6	80.2	79.2	78.4	78.3	78.2	78.0	77.2	75.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	79.9	81.0	82.2	82.7	82.0	80.6	80.3	80.3	80.1	80.0	79.3
		MCDB	85.4	87.2	88.8	89.2	87.8	86.5	85.9	85.4	85.7	85.7	86.0	85.2
	2%	WB	77.8	79.0	80.1	81.4	81.9	80.8	79.8	79.5	79.6	79.5	79.1	78.5
		MCDB	84.6	85.8	87.6	88.5	87.2	85.8	85.0	84.5	85.0	84.9	85.0	84.5
	5%	WB	77.0	78.2	79.5	80.6	81.2	80.1	79.2	79.0	79.0	79.0	78.5	77.7
		MCDB	84.1	85.2	87.0	87.9	86.9	85.2	84.2	84.1	84.6	84.4	84.3	83.9
	10%	WB	76.2	77.4	79.0	80.0	80.6	79.7	78.8	78.6	78.6	78.5	77.9	77.0
		MCDB	83.1	84.7	86.3	87.0	86.6	84.7	83.8	83.7	84.1	83.9	83.7	83.2
Mean Daily Temperature Range	5% DB	MDBR	15.4	15.5	13.6	12.1	8.7	7.1	6.6	7.0	8.0	9.6	11.6	12.9
		MCDBR	17.6	17.8	14.8	12.9	9.8	7.5	7.1	7.6	8.7	10.5	13.7	15.2
		MCWBR	5.8	6.2	5.9	5.4	4.3	3.8	3.3	3.3	3.8	4.5	5.3	5.4
	5% WB	MCDBR	13.9	14.3	13.4	12.1	8.9	7.5	6.9	7.4	8.2	9.8	11.7	12.3
		MCWBR	5.6	6.2	5.9	5.6	4.5	4.0	3.6	3.5	4.1	4.6	5.2	5.4
Clear-Sky Solar Irradiance	taub	0.432	0.444	0.489	0.501	0.522	0.500	0.515	0.497	0.466	0.474	0.465	0.452	
	taud	2.374	2.322	2.212	2.205	2.134	2.209	2.142	2.212	2.314	2.292	2.316	2.336	
	Ebn at Noon	280	280	267	260	249	251	248	258	270	268	268	272	
	Edh at Noon	39	43	48	47	49	45	49	46	43	43	41	40	
All-Sky Solar Radiation	RadAvg	1664	1889	2009	1902	1721	1699	1742	1776	1795	1700	1489	1477	
	RadStd	124	101	89	92	126	91	85	85	115	99	112	128	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page